

OE-EC803A — Internet of Things

Semester VIII • 3L:0T:0P • 3 Credits

Module 3: Prototyping: Thinking & Embedded Hardware Platforms

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1. Thinking About Prototyping

Prototyping is the iterative process of quickly assembling a working model of a device to test assumptions and functionality before entering mass manufacturing.

1.1 Prototyping Considerations

- **Sketching:** Fast, non-functional drawings to visualize hardware forms and user flows.
- **Familiarity:** Choosing tools the designer already knows (e.g. Arduino IDE) to reduce development time, even if other tools are mathematically more optimal.
- **Costs vs. Ease:** Fast prototyping tools (like jumper wires and breadboards) are cheap and easy, but produce bulky, fragile systems. Production-ready designs require custom PCBs.
- **Open Source vs. Closed Source:** Open hardware/software allows developers to adapt existing code libraries and hardware schematics, tapping into massive global communities.

2. Basic Electronics for IoT

IoT edge nodes interface directly with the physical world using sensors and actuators.

2.1 Basic Quantities

Ohm's Law and Power

Ohm's Law relates voltage (V), current (I), and resistance (R):

$$V = I \cdot R \quad (1)$$

Electrical power (P) in watts is given by:

$$P = V \cdot I = I^2 \cdot R \quad (2)$$

2.2 Signals and Interface Elements

- **Digital vs. Analog:** Digital signals are binary (0 or 1, representing 0V and 3.3V/5V). Analog signals vary continuously (e.g., LDR resistance changing with light).
- **PWM (Pulse Width Modulation):** Mimics analog outputs from a digital pin by switching the pin ON and OFF very rapidly. The ratio of ON time to the period is called the **Duty Cycle** (D):

$$V_{\text{eff}} = D \cdot V_{\text{max}} \quad (3)$$

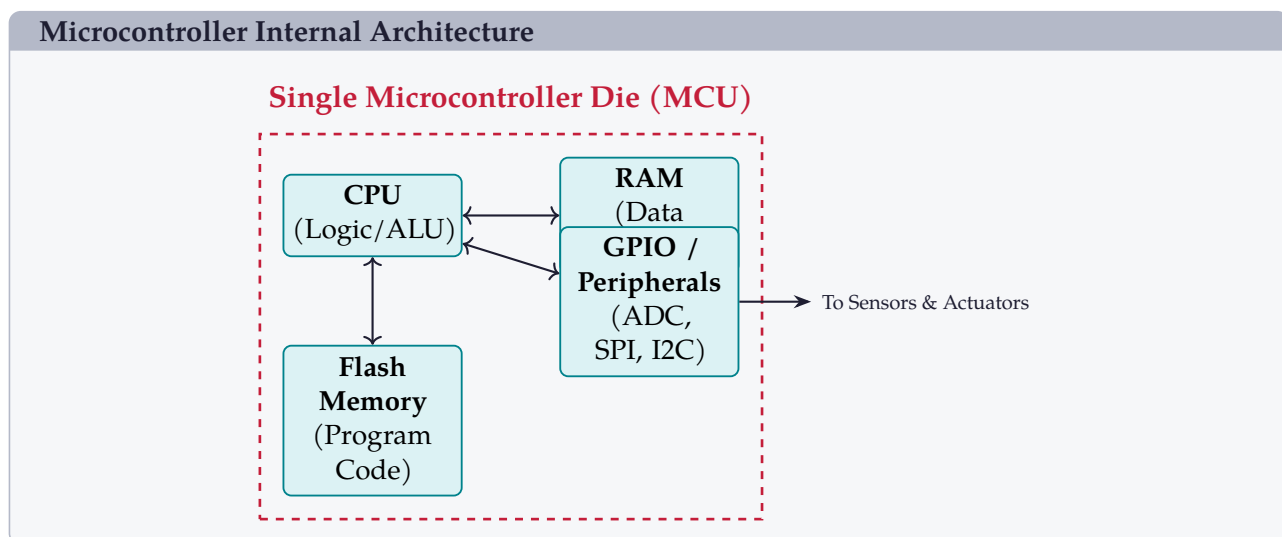
- **DHT11 Sensor:** A digital sensor measuring temperature and humidity, sending data over a single-wire digital protocol.
- **Relay:** An electromagnetic switch allowing a low-power microcontroller pin to switch high-voltage/high-current AC devices.

3. Prototyping Embedded Platforms

IoT designs select hardware platforms based on processing power, OS requirements, and connectivity.

3.1 Embedded Computing Basics

- **Microcontroller (MCU):** A single integrated circuit containing a CPU, RAM, Flash memory, and I/O peripherals (GPIO, ADC, SPI, I2C). Low-power, real-time execution. Example: ATmega328P.
- **Microprocessor (MPU):** Contains only the CPU. Requires external RAM, Flash, and support ICs. Runs complex Operating Systems (Linux). High power consumption. Example: ARM Cortex-A72.



3.2 Hardware Platforms Overview

1. **Arduino:** Open-source microcontroller board. Ideal for real-time bare-metal control, low power states, and simple sensor interfacing. Programmed in C++ using the Arduino IDE.
2. **Raspberry Pi:** A Single-Board Computer (SBC) running a full Linux OS. Features built-in Wi-Fi, Bluetooth, HDMI, and high computing power. Handles edge gateway routing and computer vision.
3. **BeagleBone Black:** An industrial-grade Linux board featuring two **PRUs (Programmable Real-Time Units)**. PRUs are independent auxiliary CPU cores that handle high-speed, deterministic real-time I/O bypass operations, making the BBB superior for precise motor control.
4. **Electric Imp:** A Wi-Fi connectivity module integrated with a cloud service. It uses the **Squirrel** programming language and automatically manages secure cloud communication and OTA firmware updates.
5. **Mobile Phone and Tablets:** High-performance consumer devices containing built-in screen displays, cellular/Wi-Fi/Bluetooth networking, and integrated sensors (GPS, accelerometer, gyroscope, camera). Ideal for rapid application-level prototyping and user interface testing.

6. **Plug Computing:** Compact, wall-plug-sized, always-on servers containing a microprocessor, RAM, storage, and Ethernet. They act as low-power home automation gateways.

Quick Revision — Module 3

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Key Formulas and Metrics at a Glance

Ohm's Law	$V = I \cdot R$ (Volts, Amperes, Ohms)
Power dissipation	$P = V \cdot I = I^2 \cdot R$ (Watts)
PWM Effective Voltage	$V_{\text{eff}} = D \cdot V_{\text{max}}$ ($D = \text{Duty Cycle} \in [0, 1]$)
ADC Resolution	$V_{\text{step}} = \frac{V_{\text{ref}}}{2^N - 1}$ (N -bit Analog-to-Digital Converter)

One-Line Definitions

- **Microcontroller:** An IC integrating CPU, memory, and peripheral interfaces on a single chip.
- **Single-Board Computer:** A complete computer built on a single circuit board, running an operating system.
- **PRU (BeagleBone):** Co-processors that run independently of the main OS to execute real-time I/O tasks.
- **PWM:** A technique to simulate analog voltages by pulsing a digital output at varying widths.
- **Relay:** An electromechanical switch used to control high-power circuits with low-power digital signals.

Frequently Asked / Viva Questions

Q1. What is the difference between a microcontroller and a microprocessor?

A microcontroller integrates the CPU, RAM, Flash memory, and I/O peripherals on a single silicon chip (System-on-Chip). A microprocessor contains only the CPU core and requires external components (RAM, Storage, Ethernet ICs) to function.

Q2. Why are microcontrollers preferred over microprocessors for simple IoT sensor nodes?

Microcontrollers consume far less power (milliwatts vs. watts), support deep sleep modes that allow battery operation for years, are much cheaper, and run code directly on bare-metal without the boot latency of an operating system.

Q3. What are the PRUs in a BeagleBone Black, and why are they useful?

The Programmable Real-Time Units (PRUs) are two auxiliary 32-bit RISC cores running at 200MHz independently of the main Linux OS. They allow BBB to

Frequently Asked / Viva Questions (contd.)

perform fast, deterministic real-time I/O tasks without scheduling delays caused by the Linux kernel.

Q4. Explain the concept of PWM and how it is used to dim an LED.

PWM (Pulse Width Modulation) switches a digital pin between high (5V) and low (0V) states rapidly. By changing the Duty Cycle (the fraction of time the pin is high), the effective average voltage varies continuously. This average voltage controls the LED brightness.

Q5. What is a Single-Board Computer (SBC)? Give an example.

A Single-Board Computer is a complete, functioning computer built on a single circuit board, incorporating a microprocessor, memory, networking, and USB ports. An example is the Raspberry Pi, which runs a Debian-based Linux OS.

Q6. How does a relay isolate a microcontroller from high-voltage appliances?

A relay uses an electromagnet. When the microcontroller energizes the internal coil (isolated low-voltage side), the magnetic field pulls a mechanical contact switch closed on the high-voltage side. There is no electrical connection between the two sides.

Q7. Why is community support considered an important parameter when choosing prototyping hardware?

IoT development moves fast. A large active community ensures open-source driver libraries are available for almost any sensor, troubleshooting forum threads exist for common bugs, and development boards are widely stocked.

Q8. What is Electric Imp, and how does it handle programming?

Electric Imp is a secure IoT module integrating Wi-Fi hardware with a managed cloud. It is programmed in the Squirrel scripting language, allowing developers to write both local device code and cloud agent code in a single framework.

Comparison Snapshot

Arduino vs. Raspberry Pi vs. BeagleBone Black:

Feature	Arduino Uno	Raspberry Pi 4	BeagleBone Black
Type	Microcontroller (MCU)	Single-Board Computer (SBC)	Industrial SBC
Processor	8-bit ATmega328P (16MHz)	64-bit Quad ARM (1.5GHz)	32-bit ARM (1GHz) + 2x PRUs
Memory	2 KB SRAM, 32 KB Flash	2GB / 4GB / 8GB LPDDR4	512MB DDR3, 4GB eMMC
Operating System	None (Bare-Metal)	Yes (Linux / Raspberry Pi OS)	Yes (Debian Linux)
Power Draw	Very Low (50mW)	Medium-High (3W to 5W)	Medium (1.5W to 2.5W)
Best Use Case	Analog sensor read, relays.	Computer vision, IoT gateway.	Real-time industrial automation.